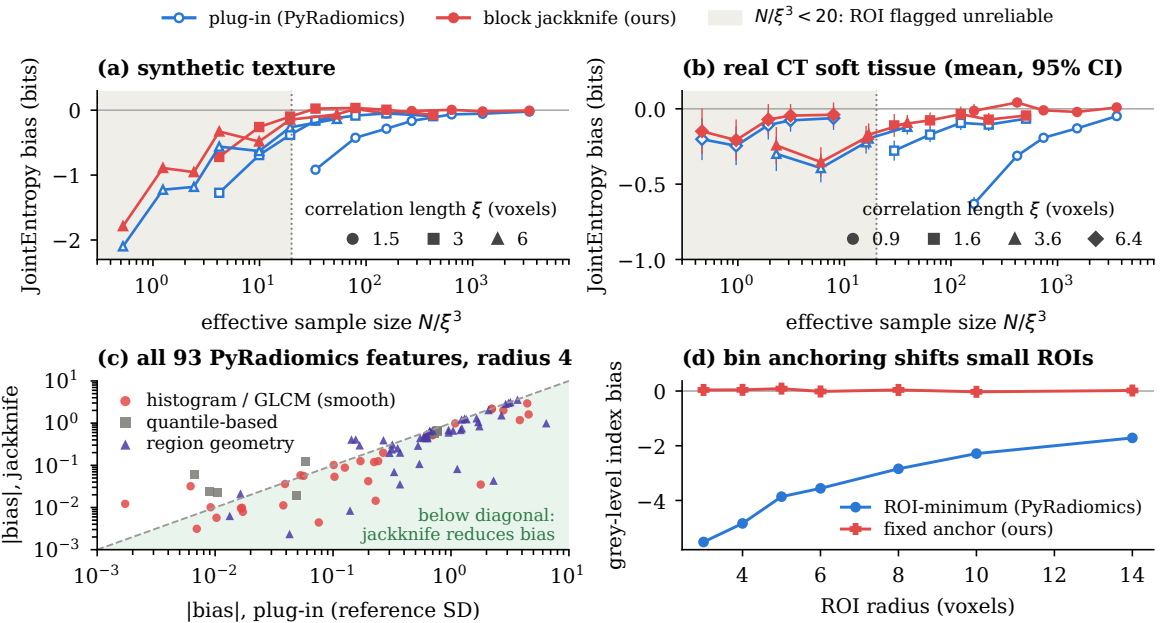
plug-in (PyRadiomics)
block jackknife (ours)
$N/\xi^3 < 20$: ROI flagged unreliable
(a) synthetic texture
JointEntropy bias (bits)
correlation length ξ (voxels)
1.5
3
6
effective sample size N/ξ^3
(b) real CT soft tissue (mean, 95% CI)
JointEntropy bias (bits)
correlation length ξ (voxels)
0.9
1.6
3.6
6.4
effective sample size N/ξ^3
(c) all 93 PyRadiomics features, radius 4
histogram / GLCM (smooth)
quantile-based
region geometry
below diagonal:
jackknife reduces bias
|bias|, jackknife
|bias|, plug-in (reference SD)
(d) bin anchoring shifts small ROIs
grey-level index bias
ROI-minimum (PyRadiomics)
fixed anchor (ours)
ROI radius (voxels)